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SUPERSEDING
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29 July 1983

MILITARY SPECIFICATION

SEMITRAILER, STAKE: TWO LEVEL DECKS, WITH HYDRAULIC LOADING RAMP, 20 TON, MODIFIED COMMERCIAL

This specification is approved for use by the US Army Tank-Automotive Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

* 1.1 Scope. This specification covers a commercial, 20 ton stake semitrailer with upper (gooseneck) and lower main decks and a tandem axle. The semitrailer is fitted with a kickup (gooseneck) deck, lower (main) deck, hydraulic rear loading ramp, hydraulic winch, and an auxiliary engine. The semitrailer is intended for transporting cargo, fork lift trucks, and fork lift trucks with cargo. Each vehicle produced under this specification is a commercial item which is warranted by the manufacturer as specified in acquisition documents.

Beneficial comments (recommendations. additions. deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Tank-Automotive Command, ATTN: AMSTA-UED, Warren, MI 48397-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A

FSC-2330

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

	MIL-P-514	- Plates, Identification, Instruction and Marking, Blank.
	MIL-W-3912	- Wood Parts, Fabricated: for Transport Vehicle Bodies and Similar Applications.
	MIL-H-46170	- Hydraulic Fluid, Rust Inhibited, Fire-Resistant, Synthetic Hydrocarbon Base.

STANDARDS

FEDERAL

*	FED-STD-297	- Rustproofing of Commercial (Nontactical) Vehicles.

MILITARY

	MIL-STD-209	- Slinging and Tiedown Provisions for Lifting and Tying Down Military Equipment.
	MIL-STD-1223	- Nontactical Wheeled Vehicles Treatment, Painting, Identification Marking, and Data Plate Standards.
*	MIL-STD-1261	- Arc Welding Procedures for Constructional Steel.
	MS 75021	- Connector, Receptacle, Electrical - 12 Contact, Intervehicular, 28-Volt, Waterproof.

* (Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, Military Specification and Standards, Bldg 4D, 700 Robbins Avenue Philadelphia, PA 19111-5094.)

* 2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

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* DEPARTMENT OF DEFENSE (DOD)
Department of Defense Index of Specifications and Standards (DODISS).

(Copies of DODISS are available on a yearly subscription basis either from the Government Printing Office for hard copy, or microfiche copies are available from the Director, Navy Publication and Printing Service Office, 700 Robbins Avenue, Philadelphia, PA 19111-5093.)

DEPARTMENT OF TRANSPORTATION (DoT)
Federal Motor Carrier Safety Regulations.
Federal Motor Vehicle Safety Standards.

* (Application for copies of DoT publications should reference the Code of Federal Regulations, 49 CFR, and the Federal Register, and should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.)

* 2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the, issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

* THE EUROPEAN TYRE AND RIM TECHNICAL ORGANISATION (ETRTO)
Standards Manual

(Application for copies of ETRTO publications should be addressed to the European Tyre and Rim Technical Organisation, 32, Avenue Brugmann, 1060 Brussels, Belgium.)

SAE, INC.
SAE Standards and Recommended Practices

*	J318	- Air Brake Gladhand Service (Control) and Emergency (Supply) Line Couplers - Trucks, Truck-Tractors and Trailers.
	J516	- Hydraulic Hose Fittings.
	J517	- Hydraulic Hose.
*	J534	- Lubrication Fittings.
	J551	- Performance Levels and Methods of Measurement of Electromagnetic Radiation from Vehicles and Devices (30-1000 MHz).
	J560	- Seven-Conductor Electrical Connector for Truck-Trailer Jumper Cable.
	J588	- Turn Signal Lamps.
	J682	- Rear Wheel Splash and Stone Throw Protection.

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	J700	- Fifth Wheel Kingpin - Commercial Trailers and Semitrailers.
	J702	- Brake and Electrical Connection Locations Truck Tractor and Truck-Trailer.
*	J706	- Rating of Winches.
	J931	- Hydraulic Power Circuit Filtration.
	J1292	- Automobile, Truck, Truck-Tractor, Trailer, and Motor Coach Wiring.

(Application for copies of SAE publications should be addressed to SAE Inc., 400 Commonwealth Drive, Warrendale, PA 15096.)

THE TIRE AND RIM ASSOCIATION, INC.
Year Book.

* (Application for copies of the Tire and Rim Association publications should be addressed to the Tire and Rim Association, Inc., 175 Montrose West Ave., Copley, OH 44321.)

* (Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

* 2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

* 3.1 Standard vehicle and accessories. Except as specified in 3.1.1 through 3.1.1.10, the semitrailer, components, assemblies, and accessories to be delivered under the contract shall be the manufacturer's standard or optional items, which meet or exceed the requirements of this specification. All vehicle items shall be as represented and rated in the vehicle and equipment manufacturer's technical data. Technical data shall be limited to specifications and technical material, identical to that furnished to the authorized company representatives for selection of vehicle models and components and shall be available in the engineering offices of the procuring activity prior to delivery of the items.

3.1.1 Social requirements. In addition to the standard semitrailer and components specified in 3.1, the vehicle shall be furnished with special equipment as specified herein.

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3.1.1.1 Treatment and painting. The vehicle shall be treated and painted in accordance with MIL-STD-1223. As specified by the procuring activity for the appropriate service (see 6.2), the exterior color shall be in accordance with MIL-STD-1223.

3.1.1.2 Marking and data plates. As specified by the procuring activity for the appropriate military service (see 6.2), identification marking and data plates shall be in accordance with MIL-STD-1223. When specified (see 6.2), concealed markings shall be furnished.

3.1.1.3 Rustproofing. When specified (see 6.2), the vehicle shall be rustproofed in accordance with FED-STD-297

3.1.1.4 Wood treatment. Wood shall be treated in accordance with MIL-STD-1223.

3.1.1.5 Additional data plates. Additional data plates depicting the maximum towing speeds, permissible loads and permissible load distributions shall be provided. The plates shall indicate the maximum towing speeds with and without payload, the maximum concentrated payload for each four foot segment of the entire length of the semitrailer, including the gooseneck, and the longitudinal location, length and width(s) of the maximum concentrated loads permissible. Appropriate loading instructions and cautions, as applicable to the manufacturer's design, shall be furnished on the plates. A plate shall be provided on each side of the vehicle in a conspicuous location. The plates shall conform to composition A (class l or 2) or composition C of MIL-P-514.

3.1.1.6 Lifting and tying down attachments. When specified (see 6.2), the semitrailer shall be equipped with lifting and tying down attachments. Lifting and tying down attachments shall conform to type II or type III of MIL-STD-209. A transportation plate conforming to composition A (class l or 2) or composition C of MIL-P-514 shall be attached to the semitrailer. The transportation plate shall be inscribed with a diagram showing the lifting attachments and proposed lifting slings, the capacity of lifting attachments and proposed lifting slings, the capacity of each attachment, and the required length and size of each sling cable. A silhouette of the semitrailer, showing the center of gravity, shall be provided on the transportation plate. Tiedown attachments shall be identified by stenciling or other suitable markings. Tiedown markings shall clearly indicate that the attachments are intended for the tiedown of the semitrailer on the carrier when shipped.

3.1.1.7 Lubrication chart. When specified (see 6.2), a lubrication chart shall be provided. The chart shall direct attention to all lubrication fittings and specify the range and grade of lubricants required for critical temperatures. The chart shall be permanently attached to the semitrailer in a readily visible location. The chart shall be inscribed on a plate conforming to composition A (class l or 2) or composition C of MIL-P-514.

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3.1.1.8 Center of gravity. When specified (see 6.2), the longitudinal center of gravity of the unloaded semitrailer shall be located and permanently indicated. The longitudinal center of gravity data shall be furnished on a photoetched or stamped corrosion resistant data plate. The plate shall be securely attached to the vehicle in a readily visible location.

3.1.1.9 Radio interference suppression. The vehicle shall be suppressed to limit electromagnetic radiation in accordance with SAE J551.

3.1.1.10 Drain plugs. All drain plugs installed throughout the vehicle for the purpose of draining lubricants shall be of the permanent magnet type.

3.2 General design.

3.2.1 Federal Motor Vehicle Safety Standards. The vehicle shall comply with all Federal Motor Vehicle Safety Standards applicable to the type of semitrailer furnished and in effect on the date of manufacture.

3.2.2 Dissimilar metals. All dissimilar metals used throughout the vehicle shall be insulated from one another to prevent galvanic or electrolytic action.

3.2.3 Ratings. Vehicle ratings shall be the manufacturer's published ratings. Component and vehicular ratings shall not be raised to meet the requirements of this specification. When published ratings are not available, verification of ratings shall be available to the appropriate engineering office of the procuring activity.

* 3.2.4 Weights. The net weight of the fully equipped semitrailer shall be the minimum practicable for the service intended. The gross vehicle weight (GVW) shall consist of the net weight and the rated payload evenly distributed over the loadspace.

* 3.2.5 Accessibility. The design of the vehicle and optional equipment shall permit access for routine servicing and shall permit access for replacement and adjustment of component parts and accessories with minimal disturbance of other components and systems.

* 3.2.6 Prohibited materials. Asbestos materials shall not be used in any form in any part of the vehicle. No item, part or assembly shall contain radioactive materials in which the specific activity is greater than 0.002 microcurie per gram or activity per item equals or exceeds 0.01 microcuries.

3.3 Performance. The fully equipped semitrailer shall be capable of transporting cargo, fork lift trucks, and fork lift trucks with cargo, with a payload weight of 40,000 pounds. The semitrailer shall have the capability of lowering and raising a rear mounted hydraulic loading ramp containing a 20,000 pound load, to and from ground position to a level load position, and of moving the cargo or lift truck to and from the loading ramp to the proper level deck position

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for transporting, while connected to or separated from the towing vehicle. The fully equipped semitrailer shall evidence no part failure, deformation, permanent set or interference between parts when towed under each of the conditions specified in Table I.

TABLE I. Performance.

Condition	Payload	Speed (minimum)
Unimproved roads; other reasonably hard, uneven terrain	None	30 mph
Unimproved roads; other reasonably hard, uneven terrain	Rated payload	10 mph
Highways	None	55 mph
Highways	Rated payload	55 mph

* 3.3.1 Turning ability. The semitrailer shall be capable of assuming a 90 degree angle to a coupled truck tractor without cramping and without damaging the semitrailer or the truck tractor.

* 3.3.2 Brake performance. The service brakes shall stop the truck tractor-semitrailer combination, with the semitrailer loaded with specified payload, within the stopping distance requirements of Federal Motor Carrier Safety Regulation 393.52. Requirements shall be met with a truck tractor loaded to its full rated gross vehicle weight (GVW) and gross combination weight (GCW).

3.3.3 Tracking ability. The semitrailer shall conform to the tracking ability requirements of Federal Motor Carrier Safety Regulation 393.70(a).

* 3.4 Payload and distribution.

* 3.4.1 Rated payload capacity. The rated payload capacity, evenly distributed over the load space, shall be not less than 40,000 pounds. The semitrailer shall be capable of supporting 20,000 pounds in any 10-foot length of loadspace. The load space is defined as the total length of the platform including both lower deck (main) and kickup (gooseneck) deck, and not less than the forward 6 feet of the hydraulic rear ramp. The semitrailer shall be suitable for the concentrated, pin point, and localized stresses caused by mobile loads being driven onto the

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vehicle, by tracks, track pads and wheels, and any other non-evenly distributed loads indicated by the full range of intended uses specified in 6.1.

3.4.2 Weight distribution. Load bearing members shall be rated for the loads imposed with the frame in a horizontal position, and the payload evenly distributed over the load space. The proportion of the gross weight of the loaded semitrailer to be supported on the axles and suspension system, measured at the ground, shall be approximately equal to, but not more than, 34,000 pounds.

3.5 Dimensions and clearances. The semitrailer shall be a two-level type having a lower main deck and a full width kickup (gooseneck) deck. The semitrailer shall conform to the dimensions specified in table II while resting on the landing legs on level ground, uncoupled from the towing vehicle, without payload. The term "lower deck" (see table II) is defined as the lower main deck and the hydraulic ramp.

TABLE II. Dimensions and clearances.

	Characteristic	Dimensions
	Overall length, feet	35 minimum, 40 maximum
	Lower deck length, feet (not less than)	29
*	Overall width, inches (plus 0, minus 1 inch)	96
	Ground clearance, inches (not less than)	10
	Landing gear ground clearance	See 3.14.1
	Turning clearance (from the centerline of kingpin to any portion of semitrailer 6 inches or more below upper fifth wheel plate), inches (not less than)	82
	Swing radius, from centerline of kingpin to most distant point on semitrailer nose, inches (not more than)	58
	Distance from centerline of kingpin to front end of semitrailer, inches (not less than)	20
*	Upper fifth wheel height (from the ground to underside of plate), inches plus 1 inch, minus 0	50
	Lower deck height, inches (not more than)	40
*	Drop height, inches (not less than)	9

* 3.6 Construction and stresses. Main frame members shall be adequately braced with a sufficient number of crossmembers to present a rigid construction with sufficient strength for the loads imposed. The outside frame members shall run the full length of the loadspace. The rear crossmembers shall extend the full width of the semitrailer. When the semitrailer is loaded with rated payload, and with the loads distributed as specified in 3.4.2, the maximum fiber stress in

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each frame member(s) shall be not greater than 50 percent of the frame material yield strength or, when specified (see 6.2), 33 percent of the frame material yield strength. Under the above cited conditions of loading, the main frame deflection shall be not greater than 2 inches. All parts of the framing shall be completely electric-arc welded. The strength of the finished welds shall be such that they will not fail or crack when the semitrailer is subjected to the loads and performance requirements specified herein.

* 3.6.1 Frame splices. Where frame members are spliced, joints shall be reinforced to fulfill the dynamic loading requirements of 3.3 through 3.4 and the static loading requirements of 3.6. Splices shall be designed to avoid stress concentration. Splice welds shall be continuous. Transition areas in the gooseneck and kickup shall be smooth and rounded to minimize stress concentrations.

* 3.6.2 Frame construction. The frame shall be of the fixed gooseneck type. The main frame rails shall be high tensile steel I-beams of not less than 50,000 psi yield strength. The kickup frame and lower deck frame shall be reinforced and welded into a single structure. The crossmembers shall be not more than 2-feet apart, except over wheels.

3.6.3 Stress analysis. The contractor shall perform a stress analysis for the semitrailer loaded as specified in 3.4.2. The stress analysis shall include shear and moment diagrams and deflection calculations. The strength of the floor, crossmembers, and outside frame members shall not be included in calculations to compute the main frame maximum fiber stress; but the weight of the floor, crossmembers, and outside frame members imposed on the main frame shall be included in the total load imposed on the main frame. Stress calculations shall include complete analysis of the gooseneck and kickup area.

* 3.7 Platform. The platform shall be of the two deck type. Each deck (lower and kickup) shall be level and completely covered having no cutouts or protrusions. The kickup deck shall be floored the same as the lower deck or may be completely covered with safety tread type steel plate with a minimum thickness of 0.187 inch, exclusive of the raised tread. The lower deck shall be floored with hardwood of not less than 1-5/16 inch uniform thickness. The lower deck wood shall be apitong, white or red oak, or white ash as specified in MIL-W-3912. Apitong is allowed for flooring provided the average weight shall be not less than 47 pounds per cubic foot (pcf), with none of the individual pieces weighing less than 42 pcf, with no more than 15 percent of moisture content at time of weighing. Laminated hardwood is not acceptable. Floorboards shall be laid lengthwise with tongue-and-groove or shiplap joints. Flooring shall be secured with corrosion-resistant screws or bolts. Not less than two securing screws or bolts shall be installed at each cross sill and not less than two on each side of each butt joint. When secured from the top, the floor shall be countersunk and the top of the countersunk screw or bolt head shall be not

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less than 0.125 inch and not more than 0.250 inch below the top of the wood surface. Butt joints shall be centered over the crossmembers and shall be staggered. Steel plate decking shall be provided over the wheel wells.

* 3.7.1 Stake pockets. The semitrailer shall have not less than fourteen side stake pockets, seven on each side. Not less than four stake pockets shall be furnished across the front and not less than four across the rear. The stake pockets shall be 4 inches square and shall be located inside the inner edge of the outer rails of the load space. Stake pockets shall be furnished with drains to prevent entrapment of water.

* 3.7.2 Lashing D rings. The semitrailer shall have fourteen lashing D rings, located as follows: two on each side of the kickup deck, four on each side of the lower deck and one on each side of the ramp. Each lashing ring and its mountings shall be capable of withstanding pulls of 10,000 pounds in any direction without permanent deformation.

3.7.3 Mounting of winch. Provisions shall be made for mounting a hydraulic winch on or under the gooseneck. Provisions shall also be made for mounting an air cooled engine driven hydraulic pumping unit on the gooseneck. The hydraulic pumping unit shall provide power to the hydraulic winch and to the hydraulic ramp.

* 3.7.4 Hydraulic rear ramp. The rear portion of the platform shall serve as a hydraulic operated ramp. The ramp shall be capable of being lowered to the ground and elevated above horizontal. The ramp shall be the full tapered type providing a single continuous plane, without slope changes or gaps. The ramp angle with respect to the ground with the rear edge of the ramp on the ground shall be not more than 18 degrees. When the ramp is lowered so that its lower rear edge is touching the ground, the upper edge shall be flush with the ground within 0.25 inch. The edge of the ramp shall be capable of being elevated to not less than 54 inches above the ground with no load on the ramp. Means shall be provided for positive mechanical locking of the loaded ramp in a horizontal travel position. The ramp shall be capable of lifting and carrying a 20,000 pound load in the level travel position with the load secured and centered on the ramp. The rear portion of the ramp shall fold under to ensure conformance to Federal Motor Carrier Safety Standard No. 108.

3.7.5 Ramp controls. Controls for operating the ramp shall be located on the left side (streetside) of the gooseneck. Ramp controls shall be within easy reach of an operator standing on the ground. All controls shall be clearly identified for their intended function, with permanently affixed photoetched or stamped identification plate(s) made of corrosion-resistant material.

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* 3.8 Hydraulic winch. When specified (see 6.2), a hydraulically operated winch shall be installed on or under the gooseneck. The winch shall conform to SAE J706. The winch shall provide not less than 9,000 pounds line pull on the bare drum. Winch line speed shall be not less than 14 feet per minute at full load. The winch shall be equipped with a drum drag brake and also with an automatic safety brake. The winch shall be wound with not less than 50 feet of 1/2 inch diameter, preformed, 6x37, improved plow steel, independent wire rope core (IWRC) rope. The rope shall be furnished with thimble and recommended crosby clips properly attached and shall have not less than four feet of chain with a 1-1/2 inch open throat hook attached.

3.8.1 Winch drum guard. A winch drum guard shall be furnished guard shall confine the cable to the area between the drum flanges.

3.8.2 Winch controls. Controls for operating the winch shall be located on the left side (streetside) of the gooseneck. The controls shall be within easy reach of an operator standing on the ground. All controls shall be clearly identified as to their intended function with permanently affixed photoetched or stamped identification plate(s), made of corrosion resistant material.

3.8.3 Remote control unit. When specified (see 6.2), in addition to the mounted winch and ramp controls specified, a two-function remote control unit with not less than 60 feet of heavy duty control cable shall be provided. The control unit shall provide independent control of the hydraulic winch and the hydraulic rear ramp. The control cable shall be wound on a reel which automatically retracts the extended cable. A reel latching mechanism shall be provided to release and restore cable tension. After latching has occurred, the cable shall remain in an extended position. Further extension of the cable shall cause the latching mechanism to release and permit the cable to retract completely when the cable is released. The remote controls shall be clearly identified for their intended function, with permanently affixed photoetched or stamped identification plate(s), made of corrosion resistant material.

3.8.4 Remote control functions. The remote control unit shall be capable of being held and operated by one hand. The activating mechanism for all controls shall be such that the controls must be held depressed for operation, with automatic return to the "off" position when the controls are released. Winch and ramp controls shall be independent of each other. The winch control unit shall provide, as a minimum, forward, reverse, and off control. The ramp control shall provide, as a minimum, up, down, and off control.

3.8.5 Hydraulic system. A self contained hydraulic pumping system shall be furnished and installed on the semitrailer gooseneck. The hydraulic system shall consist of an air-cooled diesel engine, hydraulic pump, hydraulic cylinders, hydraulic fluid reservoir, control and relief valves,

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filters, hoses, tubing, fittings, and all other components required to actuate and control the hydraulic powered winch and ramp. The hydraulic system shall have the capability of furnishing not less than 17 gallons per minute (gpm) flow rate under all operating load conditions.

3.8.5.1 Hydraulic pump. The hydraulic pump shall be capable of delivering a flow rate at maximum system operating pressure which is adequate to provide for the minimum system flow rate specified in 3.8.5 and the winch line speed specified in 3.8.

3.8.5.2 Hydraulic reservoir. The hydraulic reservoir shall have sufficient capacity to hold the maximum volume of fluid required for the system plus a minimum 20 percent reserve. Reservoir minimum capacity shall be four gallons. The reservoir shall be vented by a separate vent or through the filler cap. The vent shall be fitted with a micronic filter element.

3.8.5.3 Hydraulic lines and fittings. All hydraulic tubing, hose, and fittings shall conform to applicable SAE standards. A minimum number of fittings, joints, and connections shall be used to prevent excessive back pressure, clogging, and leakage. All hoses and fittings shall be capable of withstanding a bursting pressure of four times the working pressure and a maximum proof pressure of at least two times the working pressure. Hydraulic hose shall be single wire braid or double wire braid, rubber covered, conforming to SAE J517. Hose fittings shall conform to SAE J516. The hydraulic system shall incorporate a filtration system conforming to SAE J931. Hoses shall be protected against chafing through utilization of supports and protective devices.

3.8.5.4 Hydraulic fluid and filter. A full-flow hydraulic filter, having a micron rating no greater than 35, shall be installed in the hydraulic system. The hydraulic fluid shall be of noncorrosive type with characteristics to withstand a temperature range from minus 40 degrees Fahrenheit (ºF) to plus 180°F. When specified (see 6.2). hydraulic fluid conforming to MIL-H-46170 shall be provided in the hydraulic system. A permanent decal located near the hydraulic system fill port shall be furnished stating "CAUTION: USE HYDRAULIC FLUID CONFORMING TO MIL-H-46170 ONLY."

* 3.8.5.5 Auxiliary engine. Unless otherwise specified (see 3.8.5.6) the hydraulic pump shall be driven by an air cooled diesel engine furnished with a 12-volt electric starter, 12 volt starting battery, battery charging alternator, voltage regulator, fuel tank with not less than a 3-gallon capacity, air cleaner, fuel gage, battery charging indicator, throttle, engine start and stop controls and all other components required to power the hydraulic system. The engine power, speed and

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torque shall provide continuous operation of the hydraulic system at the flow rate specified (see 3.8.5), the pressure required to operate the winch (see 3.8) under maximum load at required speeds, and operation of the ramp under maximum load.

3.8.5.5.1 Engine sound level. The engine noise level during winch and ramp operation shall be not greater than 84 dB(A). The noise level shall be measured at the operator's station adjacent to the mounted winch and ramp controls, or at the loudest point within reach of the remote control cable with the operator standing on the ground.

* 3.8.5.6 Rechargeable electrical system. When specified (see 6.2), a rechargeable electrical system shall be furnished to drive the hydraulic ramp (and winch), in lieu of an auxiliary engine. The electric motor shall be mounted on the upper gooseneck in conjunction with the hydraulic pump. The electrical system shall include:

(a) 36 volt dc motor
(b) 375 ampere-hour industrial battery
(c) Battery charging indicator
(d) Pressure switches and control circuits
(e) Connections and cables for recharging the battery from a coupled truck tractor
(f) A weatherproof battery box.

* 3.9 Stake racks. When specified (see 6.2), stake racks shall be furnished. Stake racks around the main deck shall be not less than 60 inches high. The kickup racks shall be of such height that the tops of the racks shall be level with the tops of main deck racks. The front rack shall be a reinforced steel panel, capable of being removed when unbolted. The front panel shall be designed to sustain an end thrust of not less than 50 pounds per square foot, when interlocked with the adjacent side racks. The number of side sections shall be as required for the stake pockets specified in 3.7.1. All sections shall be easily removable without evidence of binding or jamming. All sections shall meet and fit so that locking devices engage and disengage freely and shall not require the use of other than small hand tools.

3.9.1 Stake rack construction. The stakes shall be either hardwood, fitted with a metal ferrule at the lower end; pressed steel not less than 10 gage (0.1345 inch) thick; or extruded aluminum of equivalent strength. When the racks are installed, the slats shall be inside the load space, with the upright posts outside. All corners shall be square. The front panel and side and rear racks shall be provided with heavy duty interlocking hardware. The rear rack sections shall be equipped with locking devices to lock the racks to the body. Slats in the side and rear racks shall be at least 13/16 by 3-5/8 inch finished dimensions. The slats in the bottom half of the racks shall be spaced as close together as practicable. The slats in the upper half shall be spread not less than 2 inches and not more than 3 inches apart. Stakes and slats, including hardware,

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shall be held together with bolts or other through-type fasteners at least 0.250 inch in diameter. All wood parts shall be hardwood (not apitong) conforming to 3.7, treated in accordance with 3.1.1.4.

3.10 Suspension system. The semitrailer shall be furnished with a tandem suspension system. Each component of the suspension system shall have a rated capacity at least equal to the load imposed, measured at the wheels at the ground, when the semitrailer is loaded with rated payload. Clearances shall prevent interference between wheels, tires, or any other part of the semitrailer, under the operating conditions specified herein.

3.10.1 Tandem axles. The semitrailer shall have tandem axles. Axle ratings shall be at least equal to the load imposed on each axle, measured at the ground, when the semitrailer is loaded with rated payload. The wheel bearings and axle spindles shall be oil lubricated. The oil viscosity shall be in accordance with the manufacturer's recommendations. The hubcaps shall have a window for visual determination of oil level. Provisions for venting, or other method of withstanding internal pressure buildup, and for replenishing the oil supply shall be incorporated. The hubs shall be fitted with seals.

* 3.11 Wheels, rims, tires and tubes. The semitrailer shall have dual wheels on each axle. Rim and tire ratings shall conform to Tire and Rim Association or European Tyre and Rim. Technical Organisation recommendations for the type and size of tires furnished. Multi-piece rims shall not be furnished. Rim sizes shall be the same for all wheels on the semitrailer. Tire size and load range (ply rating) shall be the same for all tires on the semitrailer. When specified (see 6.2), disc type wheels shall be furnished.

* 3.11.1 Tires. Tires shall be tubeless type with highway tread. Tires shall be of the steel belted radial type. Tires shall have a rated capacity at least equal to the load imposed on each tire, measured at each wheel at the ground, with the semitrailer loaded with rated payload. Tires shall conform to Tire and Rim Association or to the European Tyre and Rim Technical Organisation recommendations.

3.11.2 Spare wheel or rim. When specified (see 6.2), a spare wheel or rim shall be mounted and secured on the carrier. When a spare tire assembly is specified, a spare wheel or rim shall be furnished.

3.11.3 Spare tire assembly. When specified (see 6.2), a spare tire assembly shall be furnished. The spare tire assembly shall include an inflated spare tire mounted on the spare wheel or rim. The spare tire shall be of the same size, load range (ply rating), and tread design as the tires installed on the semitrailer.

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* 3.11.4 Carrier for snare tire assembly. When specified (see 6.2), a carrier for a spare wheel or rim and tire assembly shall be installed in a readily accessible location on the vehicle. Threaded fasteners, when used to secure the spare tire in the carrier, shall be constructed of or plated with corrosion-resistant material. The carrier design shall enable safe removal or mounting of a spare wheel assembly using only the tools specified in 3.20. The carrier shall enable the safe removal and installation of the spare tire assembly from and to the vehicle and carrier without personnel positioning themselves or any part of their body under the spare tire assembly.

3.12 Service brakes. The brakes shall be of the full-air, internal expanding type and shall conform to Federal Motor Carrier Safety Regulations 393.40, 393.42, 393.43 and 393.45 through 393.47. The braking system shall include standard breakaway features, relay emergency valve, air reservoirs, automatic slack adjusters, piping, hose connections, gladhands, spring loaded dust covers or dummy gladhands equipped with security chains, and all other components required for a complete air brake system. The gladhands shall conform to SAE J318. The location of the gladhands shall comply with SAE J702. The braking system shall be installed in a manner which provides road clearance for travel over uneven terrain and protection against damage caused by objects striking the components. No part of the braking system shall extend below the bottom of the wheel rims. The air reservoirs shall be of sufficient capacity to supply air for the air brakes and the complete control system of the semitrailer, with an adequate reserve capacity.

* 3.12.1 Parking brakes. The semitrailer shall be equipped with spring or air diaphragm mechanical lock type parking brakes. The parking brakes shall hold the semitrailer, with rated payload, on a 10 percent grade despite the depletion of the compressed air supply. With no additional application, the parking brakes shall hold the semitrailer, with rated payload, on a 10 percent incline with the semitrailer headed first up and then down the incline. The parking brake shall be automatically applied upon disconnection of the emergency air line and under emergency conditions. The parking brakes shall conform to Federal Motor Carrier Safety Regulation 393.41.

* 3.13 Upper fifth wheel. The upper fifth wheel plate shall completely cover a 36 inch diameter lower fifth wheel when the lower fifth wheel is coupled to the kingpin of the upper fifth wheel and turned at all angles with the semitrailer. The upper fifth wheel shall conform to Federal Motor Carrier Safety Regulation 393.70(b).

* 3.13.1 Upper fifth wheel plate. The upper fifth wheel plate shall be not less than 0.375 inch thick. It shall be reinforced on the upper surface through the use of commercial steel angles, channels, and ribs and shall include an access hole to permit servicing the brake connections and coupling sockets.

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* 3.13.2 Pickup plate. A turned up lip pickup plate, compatible with the size and weight of the semitrailer, shall be provided as an extension of the upper fifth wheel plate. The pickup plate shall have a turned up lip not less than 2 inches high. The pickup plate shall extend sufficiently forward to protect the semitrailer from damage when coupling to and uncoupling from a truck tractor. The pickup plate shall be integral with the upper fifth wheel plate.

* 3.13.3 Kingpin. A 2-inch diameter kingpin conforming to SAE J700 shall be furnished. The kingpin shall be fabricated of heat treated alloy steel.

* 3.14 Landing gear. The semitrailer shall have two vertical lift, telescopic, nonrotating landing legs, with two-speed gears and a handcrank on the curbside. The landing legs shall be equipped with steel wheels or self-leveling skid pads. Supports for the crank extension shaft and clips for holding the crank when folded shall be provided. The landing gear shall withstand, without deformation, the combined static and dynamic forces due to the proportion of gross weight sustained and the forces resulting from impact during coupling and uncoupling operations. The landing gear assembly shall be mounted to plates that are gusseted to resist forward and rearward dynamic thrust on the landing legs. When placed in travel position, the landing gear legs shall remain positively locked. The landing gear shall be protected to prevent the entrance of foreign matter which would impair its functional or mechanical efficiency. The maximum cranking effort to raise the fully loaded semitrailer shall be not greater than 100 pounds.

* 3.14.1 Landing gear range of adjustment. The landing gear shall have a range of adjustment sufficient to vary the height of the upper fifth wheel from 41 inches to not less than 53 inches from the ground. With the semitrailer coupled to a towing tractor and in level position, the clearance under the fully retracted landing gear shall exceed the semitrailer ground clearance, but shall in no case be less than 11 inches.

* 3.15 Lubrication system. All moving parts normally requiring lubrication shall be provided with accessible pressure grease fittings. Pressure type lubrication fittings shall be readily accessible for servicing and shall conform to SAE J534. Pressure relief fittings shall be used where high pressure lubricating-equipment could cause damage to seals.

* 3.16 Lighting. All lights and reflectors shall be protected from operational hazards by mounting in recessed or otherwise guarded locations. The brake lights shall override the four-way emergency flasher or the two systems shall be independent of each other. Lights and reflectors shall not be mounted on vertical surfaces of the rub rails (unless recessed and fully protected), or on the semitrailer bumper. Turn signal lamps shall conform to SAE J588. All electrical wiring shall conform to SAE J1292. Clearance and identification lamps shall be constructed for easy removal and replacement of lamps and lenses without the use of hand tools.

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3.16.1 Twelve-volt direct current (dc) system. A 12-volt dc lighting system conforming to 3.16 and 3.16.2 shall be provided. The 12-volt dc lighting system shall conform to Federal Motor Carrier Safety Regulations 393.14, 393.20, 393.22, 393.23, 393.25 through 393.29, 393.32 and 393.33.

3.16.2 Receptacle for 12-volt dc system. The front of the semitrailer shall be equipped with a 7-contact receptacle conforming to SAE J560 with the connectors connected and color coded as specified therein. The receptacle shall be provided with a spring loaded cover. The receptacle shall be located in accordance with SAE J702.

3.16.3 Independent 24-volt dc system. When specified (see 6.2), a 24-volt dc lighting system, independent of the 12-volt dc lighting system, shall be provided. Separate wiring and bulbs shall be provided for the 24-volt dc system. The system shall conform to 3.16. The 24-volt dc system shall also conform to the Federal Motor Carrier Safety Regulations referenced in 3.16.1, except the 24-volt dc stop and turn lamps shall be combined.

3.16.4 Interconnected 24-volt dc system. When specified (see 6.2), capability of operating the 12-volt do system (see 3.16.1) from a towing vehicle equipped with a nominal 24-volt dc electrical system shall be provided.

3.16.5 Receptacle far 24-volt dc system. When either an independent or interconnected 24-volt dc system is specified, the front of the semitrailer shall also be equipped with a 12-contact receptacle and cover conforming to MS 75021, part number MS 75021-1. The receptacle shall be located in accordance with SAE J702. The 24-volt dc, 12-contact receptacle shall be connected to the lights as follows:

Contact B.	Connect to left-hand turn signal and stop lamp (yellow).
Contact D.	Connect to ground (white).
Contact E.	Connect to clearance, side marker, identification, and tail lamps (black and brown).
Contact J.	Connect to right-hand turn signal and stop lamp (green).
Contact L.	Connect to ground (white).

The remaining contacts shall not be connected. Circuits B and J on tactical (military design) trucks are combination stop and turn indicator circuits. On the interconnected 24-volt dc system, the normal 12-volt dc turn signal lights will function both as turn signals and stop lights, and the normal 12-volt dc stop lights will not be operational when the semitrailer is connected to a towing vehicle with a 24-volt dc power supply. Because of this condition, the stop light (red) circuit is not connected to the 24-volt, 12-contact receptacle.

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* 3.16.6 Resistors. When an interconnected 24-volt dc system is specified (see 3.16.4), the 12-contact receptacle shall be provided with resistance to each circuit to reduce the voltage of the tactical towing vehicle from a nominal 24 volts dc to 12 volts dc. Each circuit resistor shall be selected to reduce the voltage of the tactical towing vehicle to within the maximum rated voltage of the semitrailer electrical components. The resistor assembly shall be located in a protective housing and provided with adequate ventilation or a heat sink to prevent overheating and any damage to resistors, wiring, or adjacent components.

* 3.17 Mud flaps. Mud flaps shall be provided to the rear of the rear wheels, in accordance with SAE J682. A metal strip not less than 0.125 inch thick and not less than one inch wide, extending the entire width of the mud flap, shall be installed to prevent the bolt heads or bolt nuts from damaging the mud flap. As an alternate method of attaching the mud flaps, tabs or clips with minimum surface contact dimensions of 1 inch high by 1.25 inch wide by 0.093 inch thick shall be furnished at each bolt.

3.18 Rear end protection. The semitrailer rear end protection shall be in accordance with Federal Motor Carrier Safety Regulation 393.86.

* 3.19 Toolbox. A tool box shall be furnished. The tool box shall provide for storage of a vehicle jack, hand tools, anti-skid chains (for outside tires on duals only) and emergency reflective triangles. Minimum dimensions shall be 22 inches by 22 inches by 12 inches. A door opening size of not less than 20 inches by 10 inches shall be furnished. The tool box shall be fabricated of not less than 12 gage (0.1046 inch) steel. The box shall be weatherproof and shall provide for locking with a padlock.

3.20 Tools. When specified (see 6.2), the semitrailer shall be furnished with tools for changing a mounted tire assembly with the spare tire assembly and shall include a hydraulic jack, jack handle and wheelnut wrench. The jack shall be of such closed height as to permit its location under axle or other satisfactory lift point at any wheel with the tire flat. The jack, without blocking, shall be capable of raising any wheel of the loaded semitrailer to a height adequate to permit removal and replacement of wheel and tire assembly.

3.21 Servicing and adjusting. Prior to acceptance of the semitrailer by the Government inspector, the contractor shall service and adjust the semitrailer for operational use including at least the following: inflation of all tires, adjustment of brakes, proper functioning of all lighting, and complete lubrication with grades of lubricants recommended for the ambient temperature at the delivery point.

* 3.22 Workmanship. Defective components or parts and assemblies which have been repaired or modified to overcome deficiencies shall not be furnished. Welded, bolted, and riveted constructed utilized shall be in accordance with the best commercial practice.

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* 3.22.1 Welding. Welding shall conform to the American Welding Society's Structural Welding code, Steel, D1.1 and, when specified (see 6.2), to the requirements of MIL-STD-1261, class 2. The surface of parts to be welded shall be free from rust, scale, paint, grease, or other foreign matter. Welding rods shall be compatible with the metal being welded. Spot, tack or intermittent welds shall not be used where structural strength is required. Weld penetration shall provide transference of maximum design stress through the base metal juncture. Fillet welds shall be provided when necessary to reduce stress concentration.

* 3.22.1.1 Welding workmanship specimens. When specified (see 6.2), the contractor shall prepare, or shall have prepared, welding workmanship specimens in accordance with the requirements of MIL-STD-1261, class 2 representing each typical weld joint encountered on the semitrailer.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements (examination and tests) as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility for ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Government verification. Quality assurance operations performed by the contractor will be subject to Government verification at unscheduled intervals. Verification will consist of observation of the operations to determine that practices, methods, and procedures of the contractor's inspection are being properly applied. Failure of the contractor to promptly correct product deficiencies discovered shall be cause for suspension of acceptance until correction has been made or until conformance of product to specification criteria has been demonstrated.

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4.3 First production vehicle inspection. The first vehicle produced under the contract shall be inspected by the contractor at his plant under the direction and in the presence of Government representatives. The purpose of the inspection shall be to determine vehicle conformity with the requirements of the contract. Weld specimens (see 3.22.4) shall be used by Government representatives as a guide in determining the quality of all applicable welds. All welds on the first production vehicle and on all subsequent production vehicles shall be compared with applicable workmanship specimens prior to priming or painting the vehicle. Acceptance of the first production vehicle shall not constitute a waiver by the Government of its rights under the provisions of the contract.

* 4.3.1 Semitrailer weight. The semitrailer shall be weighed to determine curb weight and distribution of curb weight on the kingpin and the rear axles. The imposed loading on the kingpin and rear axles, with payload, shall be computed using curb weight and the semitrailer payload distributed as specified in 3.4 thru 3.4.2. The rated capacity of the axles, suspension system and tires shall be compared to the loads imposed to determined if they conform to contract requirements.

* 4.3.2 Road test. The semitrailer, loaded with rated payload, shall be coupled to a truck-tractor. The tractor-semitrailer combination shall be driven a distance of not less than 100 miles under the conditions specified under 3.3. At least 20 percent of the distance shall be over hard uneven terrain. At least five sudden stops shall be made from a speed not less than 20 mph. Tracking ability shall be observed to verify conformance to 3.3.3.

* 4.3.2.1 Test failure. The vehicle shall successfully complete the entire test. Failure of the vehicle to successfully complete the test shall be cause for nonacceptance of any of the contract quantity, pending correction of deficiencies and evidence of corrective action which shall prevent the recurrence of similar, deficiencies in the contract quantity. Failure of the vehicle to successfully complete the test shall not constitute an excusable delay in meeting scheduled deliveries. The equipment shall be closely monitored throughout the test for interference with towing operations, and for instability as evidenced by poor tracking or uncontrollable sideslipping, skidding, swerving, or tilting. The equipment also shall be closely monitored for loosening of parts, interference between parts, leakage of fluids or lubricants, overheating of components, and for damage, permanent distortion, or excessive wear of parts and components. The equipment shall be disassembled to the extent necessary for inspection for unusual wear or damage to components. Substitution of a new semitrailer or replacement of a major component may require complete retest at the discretion of the Government representatives. Rejection of the test semitrailer shall be for damage or deficiencies, including but not limited to the following:

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(a) Damage caused by collision
(b) Failure of any major component
(c) Vibration due to misalinement of wheels or frame
(d) Vibration due to the type of construction or mounting
(e) Evidence of abnormal tire wear due to misalinement or unbalance
(f) Failure of any vehicular safety device such as brakes or electrical circuits
(g) Evidence of structural weakness in any part of the vehicles, vehicle components, accessories or welds
(h) Loose mountings of parts or accessories due to workmanship or vehicular operation.

4.3.3 Turning ability test. A truck-tractor shall be coupled to the semitrailer and driven through turns up to 900 to the towed vehicle, right and left to determine conformance to 3.3.1. At the conclusion of five turns in each direction, the tractor and the trailer shall be examined for evidence of binding, misalinement or other damage.

4.3.4 Brake tests. The semitrailer, carrying the rated payload and coupled to the truck-tractor, shall be driven on smooth, dry, level pavement and brakes applied from a speed of 20 mph, to determine conformance to the stopping distance required under 3.3.2. Breakaway situation shall be simulated to verify braking ability upon breakaway.

4.3.5 Parking brake test. To test parking brakes, the semitrailer, with rated payload, shall be placed on a 10 percent grade, first headed up the grade and then headed down the grade, and tested to determine conformance to 3.12.1. These tests shall be conducted with the semitrailer disconnected from the towing vehicle and resting on the landing legs.

4.3.6 Landing leg test. With the trailer disconnected from the towing vehicle and loaded with the rated payload, the landing legs shall be raised and lowered to vary the height of the upper fifth wheel from 47 inches to not less than 52 inches from the ground, and tested to determine conformance to 3.14. The trailer with the rated payload shall be coupled and uncoupled no less than 5 times and the landing legs shall be examined for binding and or misalinement. Ground clearance when coupled to the towing vehicle shall be no less than 11 inches below the lowest point of the landing legs.

4.3.7 Ramp operation test. Using the semitrailer mounted controls, the ramp, without payload, shall be lowered to the ground; elevated so that the loading edge of the ramp is not less than 54 inches above ground; and lowered and secured in the horizontal travel position. Ramp operation shall conform to the requirements specified in 3.7.4 when the ramp is being powered by the ramp hydraulic system specified in 3.8.5. The test shall be conducted five times with mounted controls and repeated five times with remote controls when furnished.

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4.3.7.1 Ramp load lifting test. A load of not less than 20,000 pounds shall be secured on the ramp with the load center of gravity positioned over the center of the ramp. The ramp shall be operated as specified in 4.3.7. Ramp operation shall conform to the requirements specified in 3.3 and 3.7.4. The test shall be conducted five times with mounted controls and repeated five times with remote controls when furnished.

4.3.8 Hydraulic winch test. The winch shall be tested for conformance to the line pull and line speed requirements as specified in 3.8 when the winch is powered by the hydraulic pumping system specified in 3.8.5. The winch cable shall be unwound from the drum until there are no overlapping strands of cable on the drum and shall be attached to a minimum of a 9,000 pound line pull load. The winch shall then be activated and shall pull the load at a speed of not less than 14 feet per minute.

4.3.9 Remote control unit.

4.3.9.1 Reel assembly. The remote control unit cable shall be extended 60 feet and retracted. Cable extension, retraction, and reel latching shall conform to the requirements in 3.8.3. The test shall be repeated for cable extensions of 25 feet, 10 feet and 5 feet.

4.3.9.2 Control operation. The remote control unit shall be used to operate the rear ramp and winch as specified in 4.3.7 and 4.3.8. Operation shall conform to the requirements specified in 3.8.3.

4.3.10 Electrical test. An electrical test shall be conducted on the first production vehicle to determine that all vehicle lights are functioning. In addition, when an interconnected 24-volt system is furnished, the circuits of the 12-contact receptacle shall be energized with not less than 28 volts dc regulated voltage for a time sufficient to stabilize the temperature of the resistor assembly for not less than 30 minutes in an ambient temperature of not less than 77°F. After the test, the resistor assembly, including resistors, wiring and adjacent components shall be examined for evidence of overheating, deterioration or damage.

4.3.11 Engine sound level test. Engine sound level shall be measured during winch and ramp tests to determine conformance to 3.8.5.5.1.

4.3.12 Stake panel test. When stake racks are specified the racks shall be examined for conformance to 3.9. The front panel shall be able to be removed and replaced when unbolted from the chassis. Side sections shall be easily removable, and shall be interchangeable, by disengaging and engaging the locking devices. The use of small hand tools is permitted. The front panel shall be subjected to an end thrust load of 50 pounds per square foot, when interlocked with the adjacent side racks, without deformation or failure of the interconnecting devices.

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4.3.13 Body treatment and painting. The certification regarding the vehicle cleaning, treating, prime painting and salt spray resistance testing, as required by MIL-STD-1223, shall be made to Government representatives at the first production vehicle inspection.

4.3.14 Production sample. Upon acceptance of the first production vehicle, it shall remain at the manufacturing facility as a production sample, and be the last vehicle shipped on the contract. The contractor shall maintain the vehicle in a serviceable condition for the duration of the contract. Release of the first vehicle for shipment prior to completion of the contract shall be at the discretion of the local Government Quality Assurance Representative (QAR).

4.3.15 Failure. Failure of the first production vehicle to meet requirements of the contract shall be cause for the Government to refuse acceptance of all vehicles under the contract until corrective action has been taken.

* 4.4 Inspection of production vehicles. The contractor's inspection system shall, as a minimum, assure that the vehicle conforms to the physical and dimensional requirements and is capable of meeting the performance requirements contained herein.

4.5 Wood treatment certification. Manufacturer's records, acceptable to the Government, shall be available to verify that all wood requiring treatment in accordance with MIL-STD-1223 has been treated.

5. PACKAGING

5.1 Vehicle processing. The vehicle shall be processed for shipment, from manufacturer's plant to initial receiving activity, in accordance with manufacturer's standard commercial practice.

6. NOTES

6.1 Intended use. The semitrailers covered by this specification are intended for general nontactical use by the Government in transporting all types of cargo, including wheeled and tracked vehicles. Cargo will be loaded on this semitrailer by forklifts having a 14,000 pound axle load supported by 7-inch wide tires.

6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
* (b) Issue of DODISS to be cited in the solicitation and, if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).

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* (c) Identification of appropriate service for painting and marking (see 3.1.1.1 and 3.1.1.2).
* (d) Concealed markings, if required (see 3.1.1.2).
(e) Rustproofing, if required (see 3.1.1.3).
(f) Lifting and tying down attachements, if required (see 3.1.1.6).
* (g) Lubrication chart, if required (see 3.1.1.7).
(h) Longitudinal center of gravity indication, if required (see 3.1.1.8).
* (i) Frame strength at 33 percent of yield strength, if required (see 3.6).
* (j) Hydraulic winch, if required (see 3.8).
(k) Remote control unit for winch and hydraulic ramp, if required (see 3.8.3).
(l) Hydraulic fluid conforming to MIL-H-46170, if required (see 3.8.5.4).
* (m) Electrical power supply for winch and ramp, if required in lieu of auxiliary engine (see 3.8.5.6).
(n) Stake racks, if required (see 3.9).
(o) Disc type wheels, if required (see 3.11).
(p) Spare wheel or rim, if required (see 3.11.3).
(q) Spare tire assembly, if required (see 3.11.4).
* (r) A carrier for a spare tire assembly, if required (see 3.11.5).
(s) Independent 24-volt lighting system, if required (see 3.11.5).
(t) Interconnected 24-volt lighting system, if required (see 3.16.4).
(u) Tools, if required (see 3.20).

6.3 Definitions.

6.3.1 Improved road. An improved road is a smooth, hard surfaced road, such as a concrete or asphalt-paved highway.

6.3.2 Unimproved road. An unimproved road is an unpaved, unstabilized road with an undulating surface having occasional chuckholes and exposed rocks.

* 6.4 Subject term (key word) listing.
Gooseneck
Hydraulic loading ramp
Hydraulic winch
Landing gear
Remote control
Resistors
Stake racks
Upper fifth wheel
Welding.

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6.5 Changes from previous issue. The margins of this specification are marked with asterisks (*) to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content regardless of the marginal notations and relationship to the previous issue.

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